



Introduction to robodk





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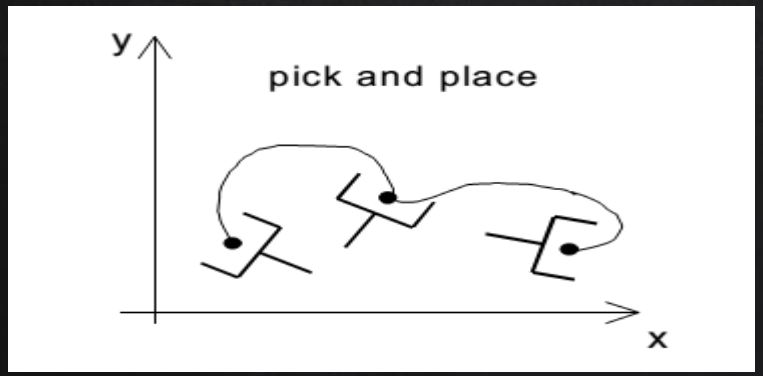


Industrial robots

Classical tasks

Pick-and-place

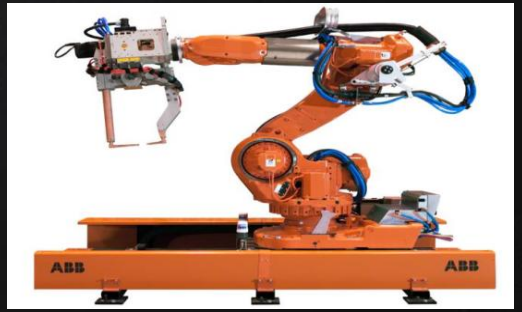




Assembling

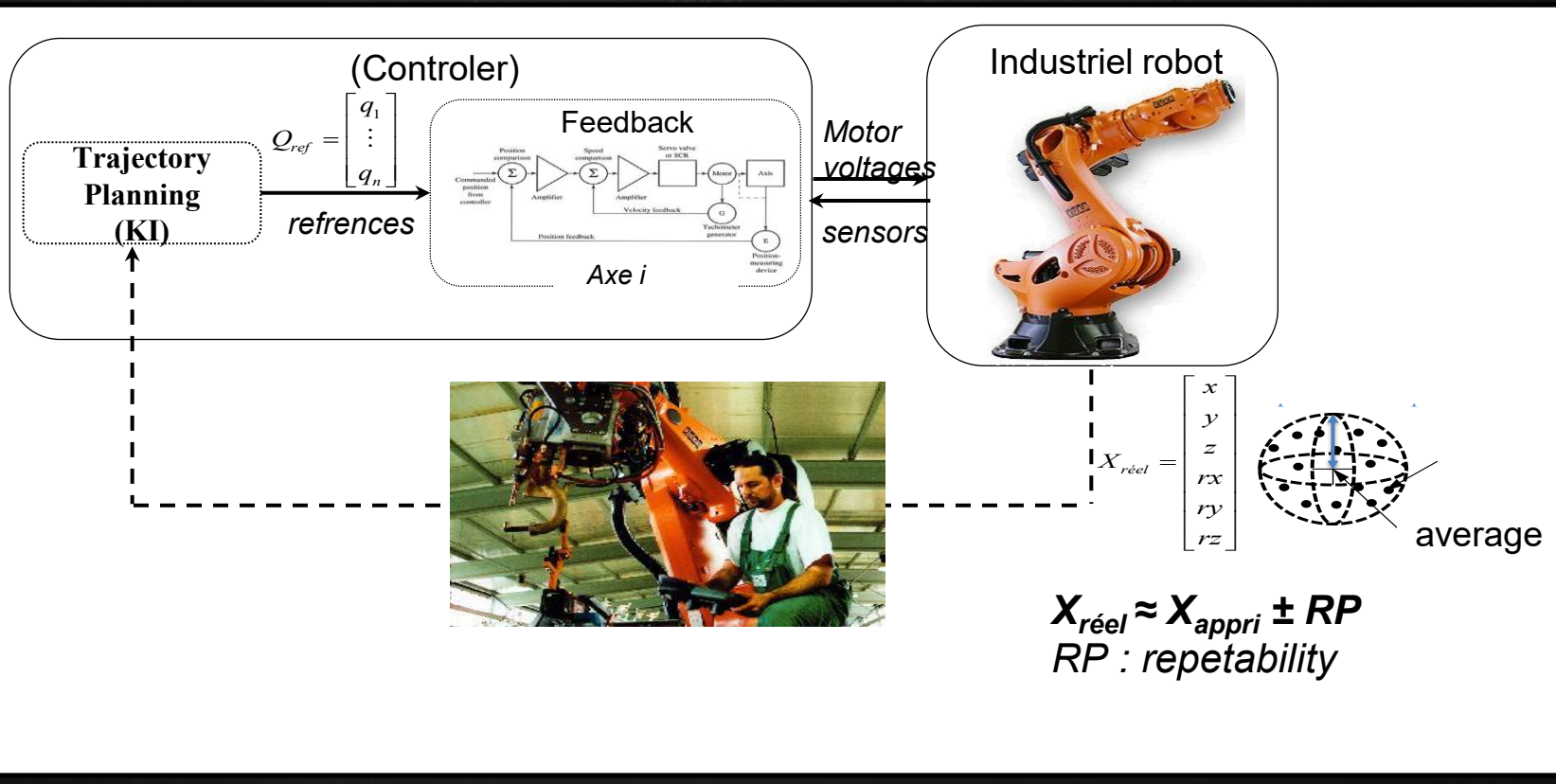


Spot welding



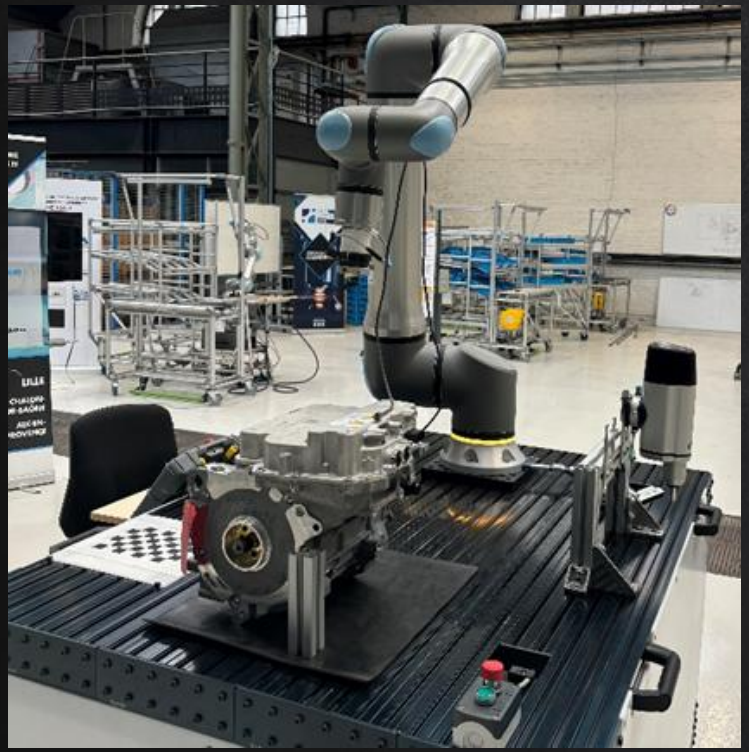
Programming mode

Teaching



Programming mode

Teaching







Programming mode

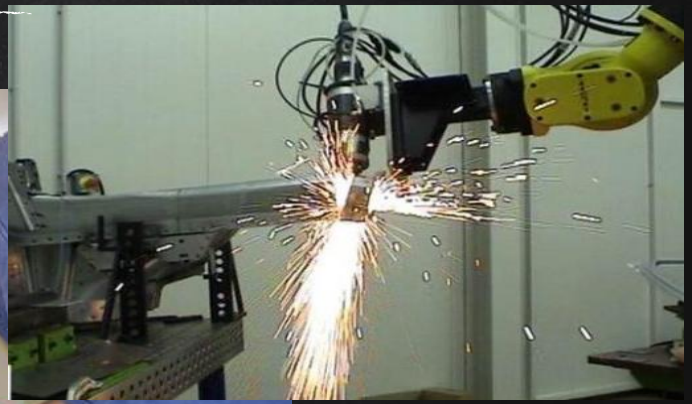
Arc welding

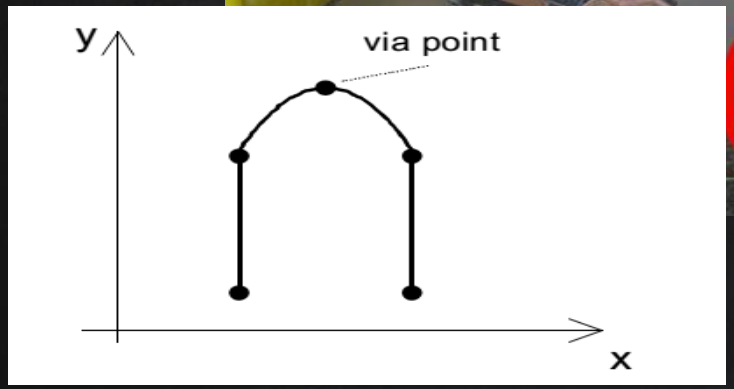


Continuous tasks



Cutting



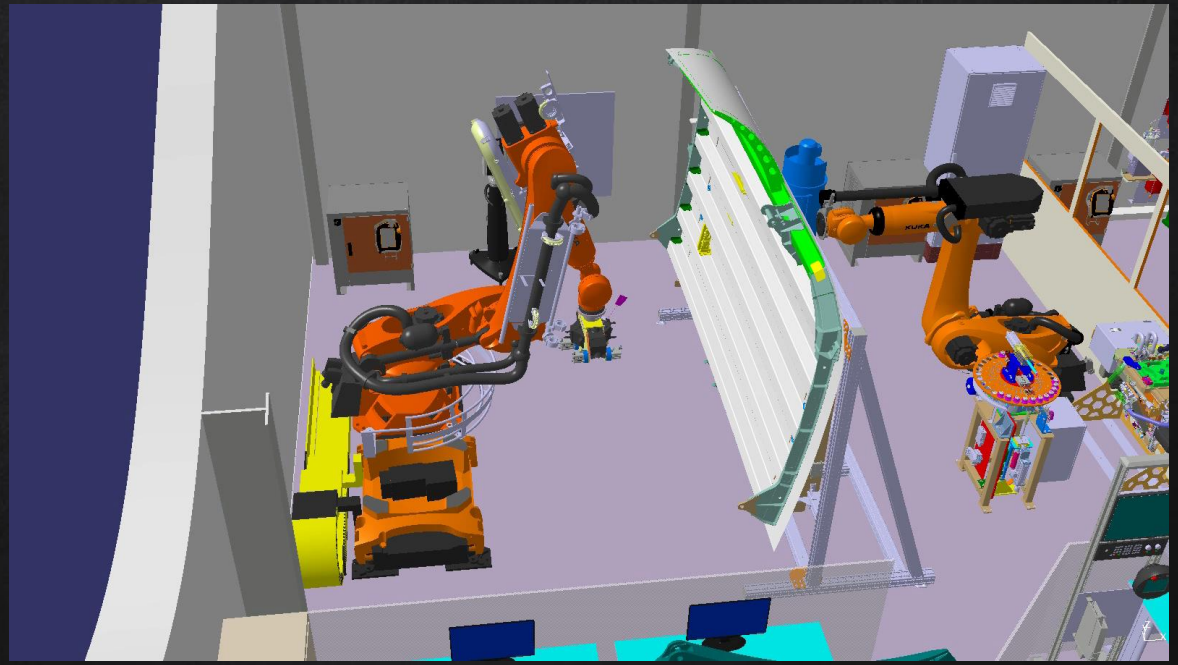


deburring



Programming mode

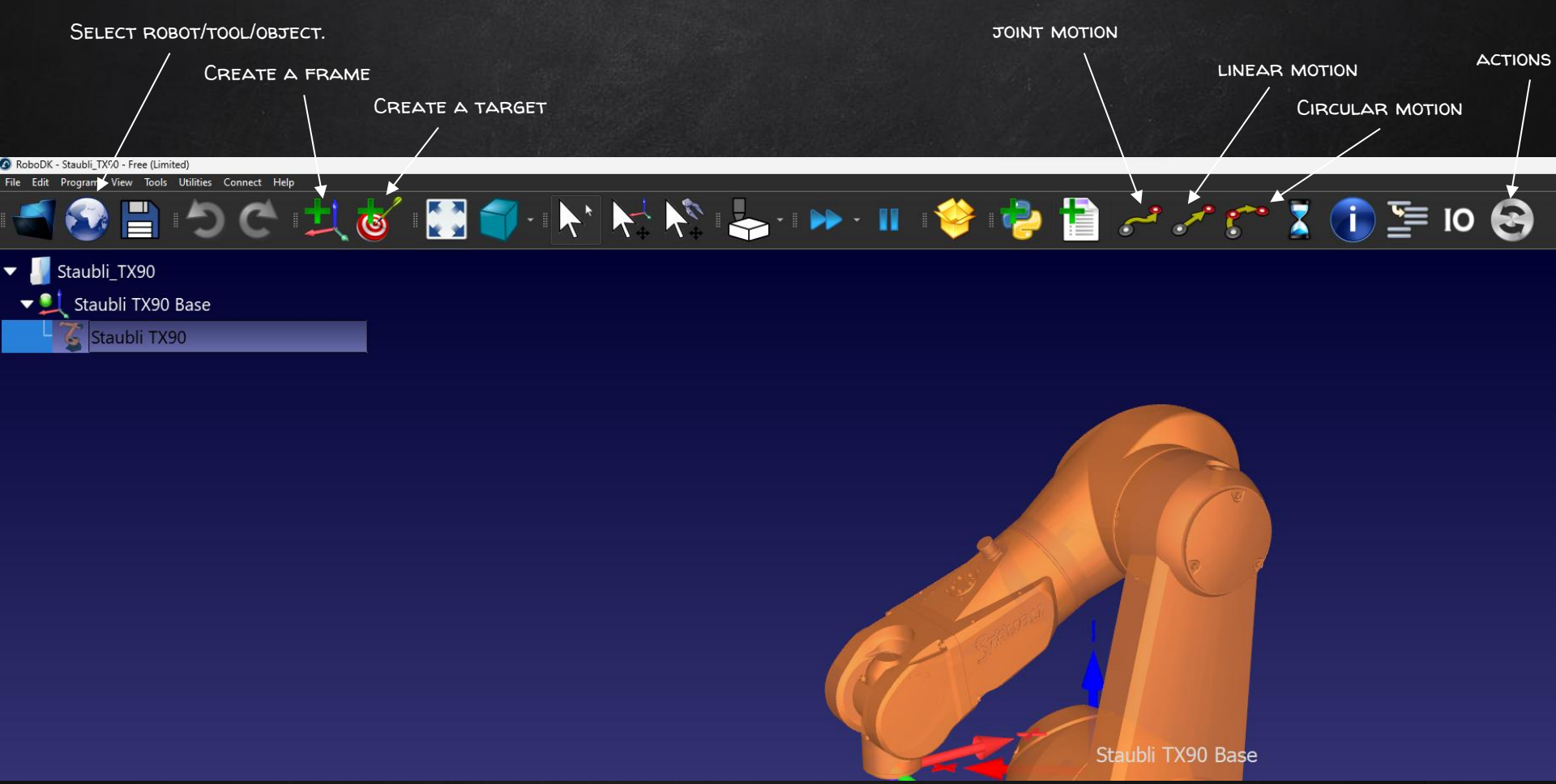
Offline programming

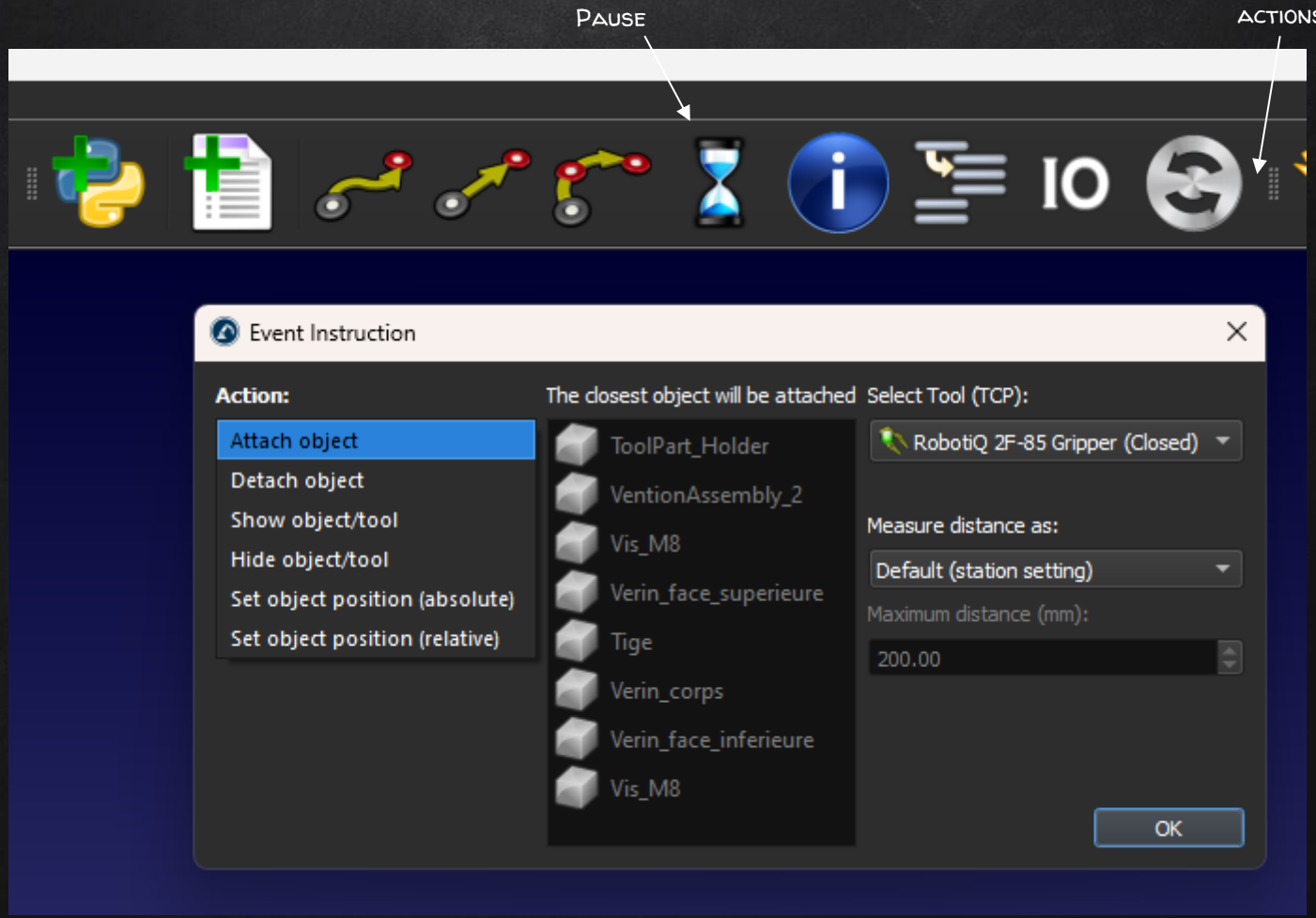


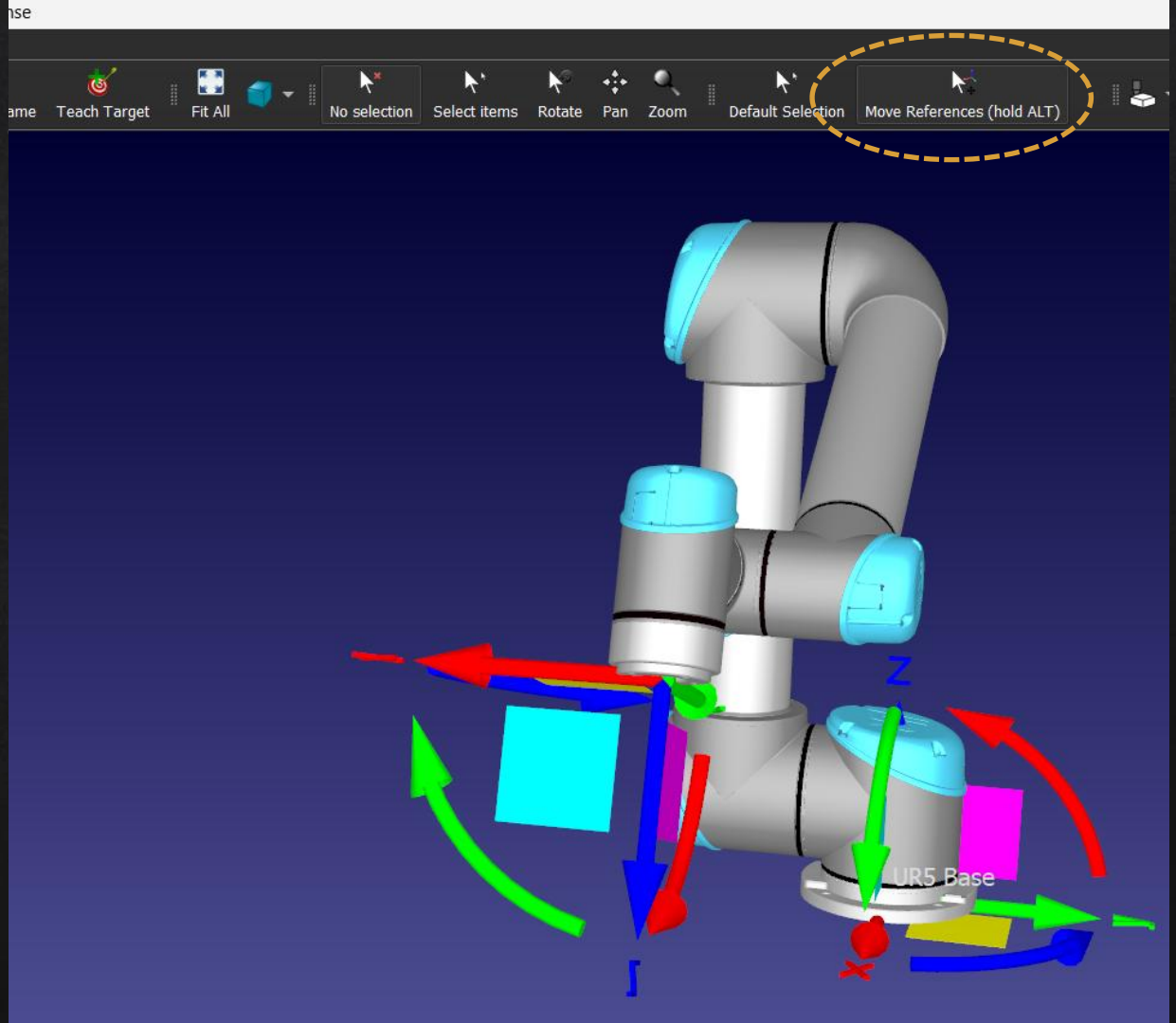
ROBOTIC ASSEMBLY OF LARGE-SCALE PARTS IN THE AEROSPACE INDUSTRY



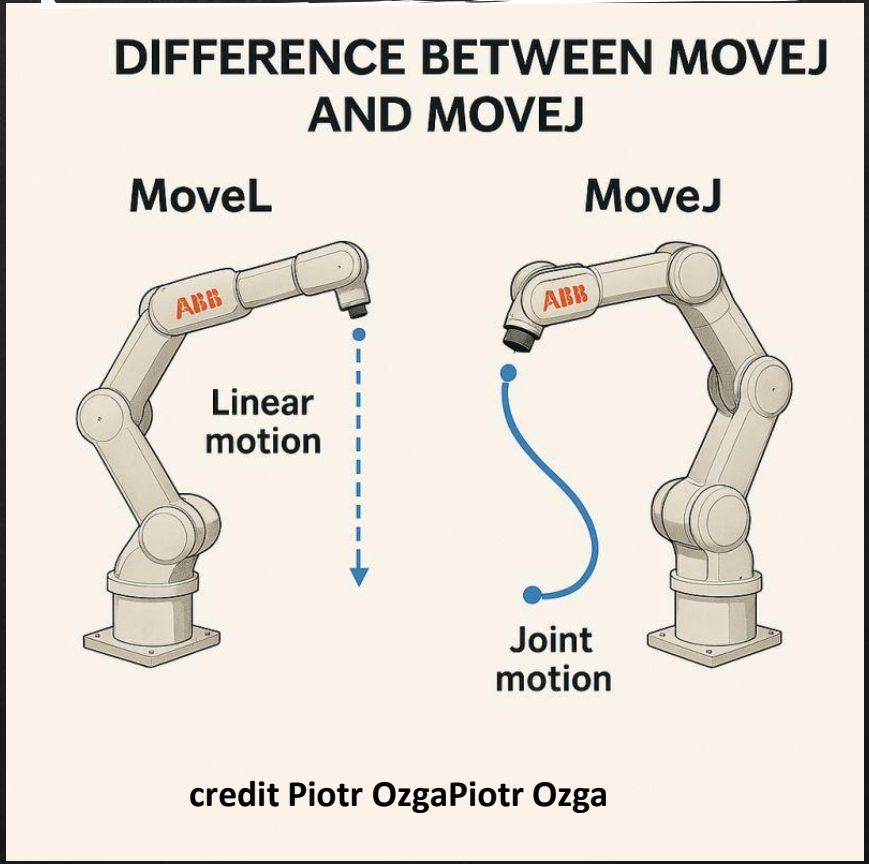
filmora 9 CRÉÉ AVEC
LE PLAN GRATUIT DE FILMORA


Select robot/tool/object.
Create a frame
Create a target
Joint motion
Linear motion
Circular motion
Actions
RoboDK - Staubli_TX90 - Free (Limited)
File Edit Program View Tools Utilities Connect Help
IO
Staubli_TX90
Staubli TX90 Base
Staubli TX90
Staubli TX90 Base


Pause
actions
IO
Event Instruction
Action:
Attach object
Detach object
Show object/tool
Hide object/tool
Set object position (absolute)
Set object position (relative)
The closest object will be attached
ToolPart_Holder
VentionAssembly_2
Vis_M8
Verin_face_superieure
Tige
Verin_corps
Verin_face_inferieure
Vis_M8
Select Tool (TCP):
RobotiQ 2F-85 Gripper (Closed)
Measure distance as:
Default (station setting)
Maximum distance (mm):
200.00
OK


Teach Target
Fit All
No selection
Select items
Rotate
Pan
Zoom
Default Selection
Move References (hold ALT)
Z
UR5 Base

Motion Types



credit Piotr OzgaPiotr Ozga

Task requirements:

- Number of degrees of freedom,
- Workspace
- Load capacity
- Speed
- Repeatability and accuracy



Robot Library

Robots | Stations | Add-ins | Posts

Search library

Clear filters | Clear search | Clear all

- Brand
- Type
- Axes
- Reach
- Payload
- Weight
- Repeatability

ABB CRB 1300-7/1.4



Brand	ABB
Model	CRB 1300-7/1.4
Axes	6
Reach	1400 mm
Payload	7 kg
Weight	79 kg
Repeatability	0.030 mm

ABB CRB 1300-10/1.15



Brand	ABB
Model	CRB 1300-10/1.15
Axes	6
Reach	1150 mm
Payload	10 kg
Weight	77 kg
Repeatability	0.023 mm

ABB CRB 1300-11/0.9



Brand	ABB
Model	CRB 1300-11/0.9
Axes	6
Reach	900 mm
Payload	11 kg
Weight	75 kg
Repeatability	0.020 mm

ABB CRB 15000



Brand	ABB
Model	CRB 15000
Axes	6
Reach	950 mm
Payload	5 kg
Weight	27 kg
Repeatability	0.050 mm

ABB IRB 120-3/0.6



Brand	ABB
Model	IRB 120-3/0.6
Axes	6
Reach	580 mm
Payload	3 kg
Weight	25 kg
Repeatability	0.010 mm

ABB IRB 140-6/0.8



Brand	ABB
Model	IRB 140-6/0.8
Axes	6
Reach	810 mm
Payload	6 kg
Weight	98 kg
Repeatability	0.030 mm

ABB IRB 580/12S



ABB IRB 580/16



ABB IRB 1100-4/0.58



ABB IRB 1100-4/0.475



ABB IRB 1200-5/0.9



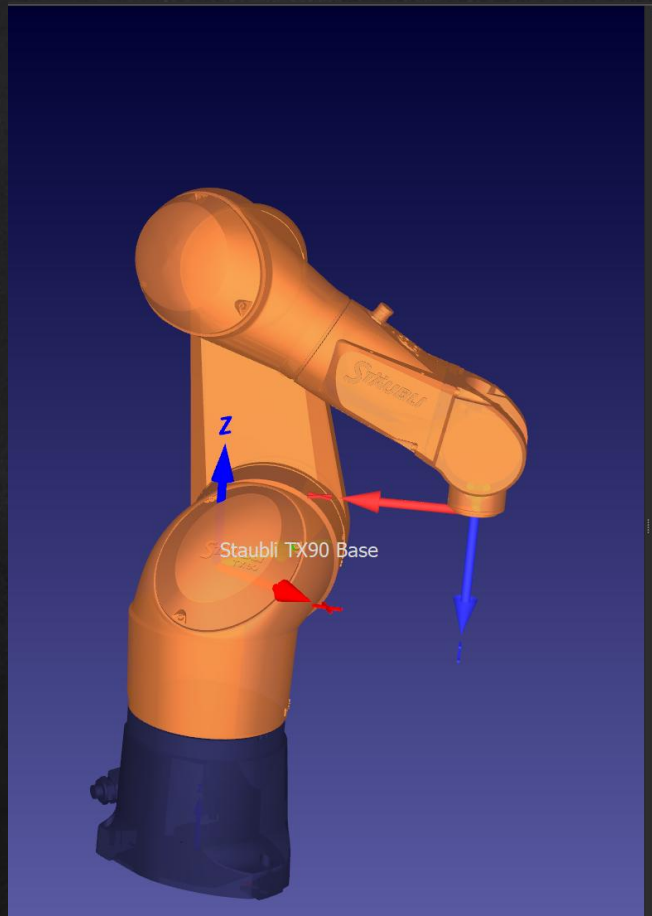
ABB IRB 1200-7/0.7

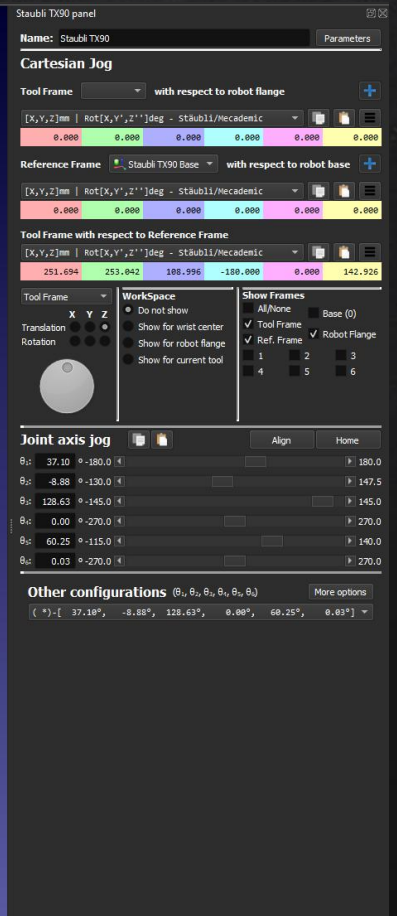


Robot Panel

Double click on robot name to open the control panel.







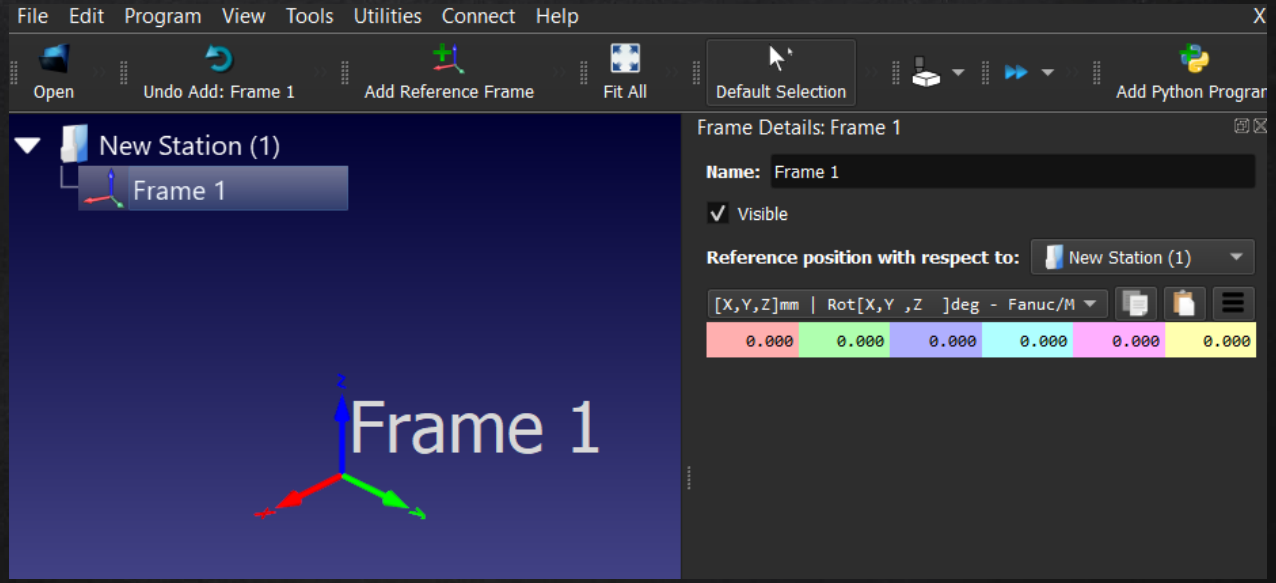
Position and orientation of "flange" with respect to robot base frame.

Attention angles method!

Joints rotations

Frames

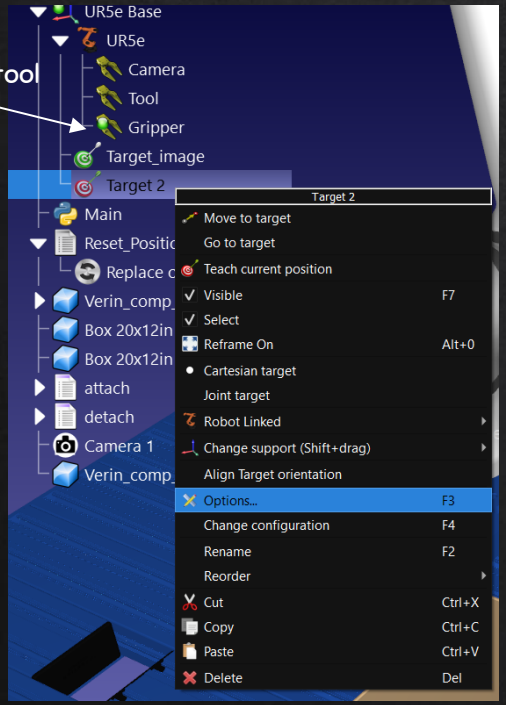
Double click on frame name.

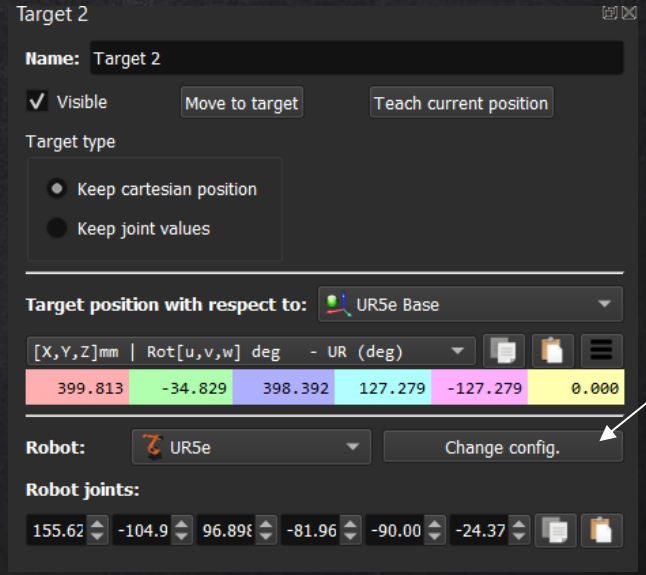


Target

Right click on target name/Options.

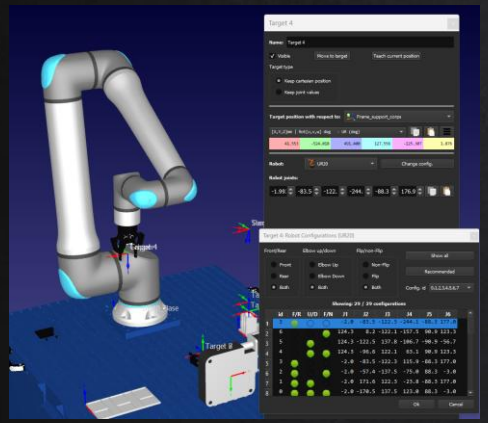
Attention to active tool





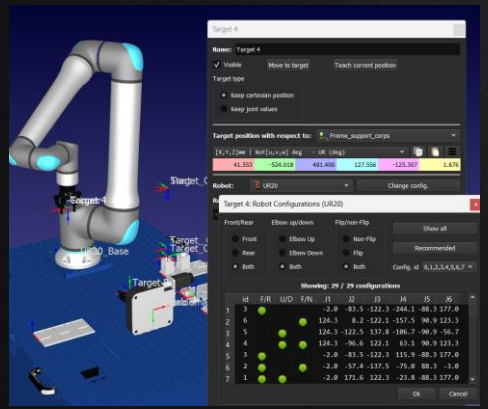
Change robot configuration

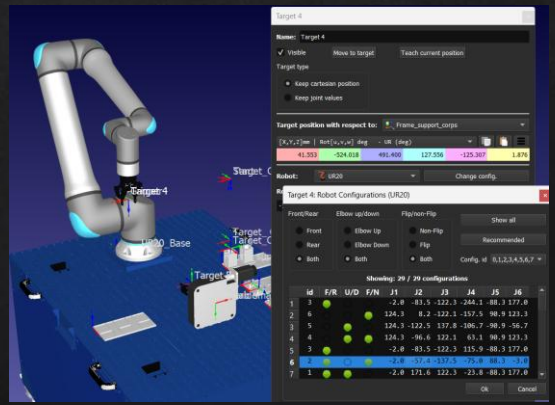
Target, configurations

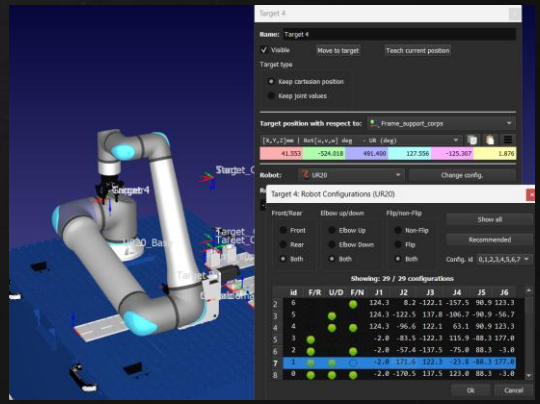








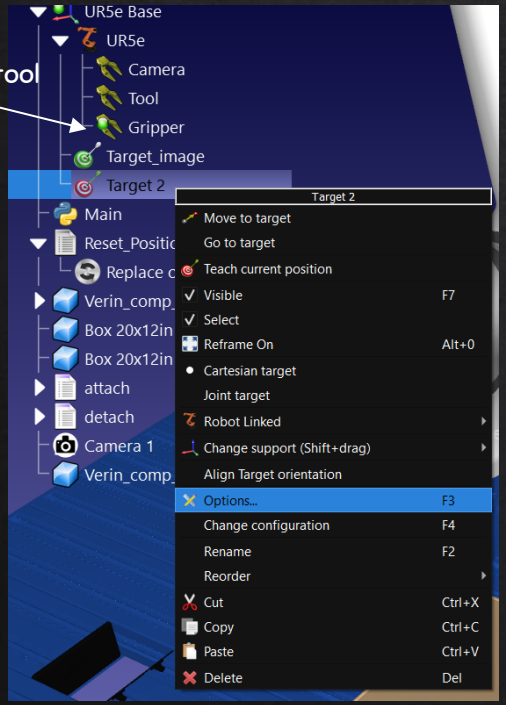


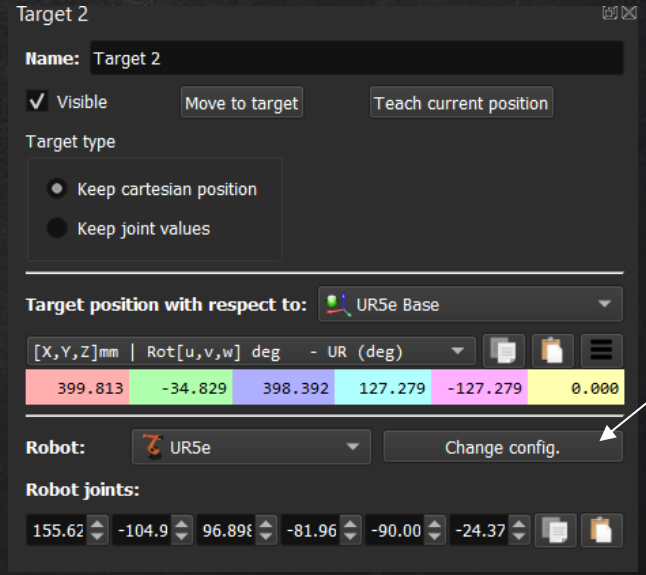


Target, Demo config

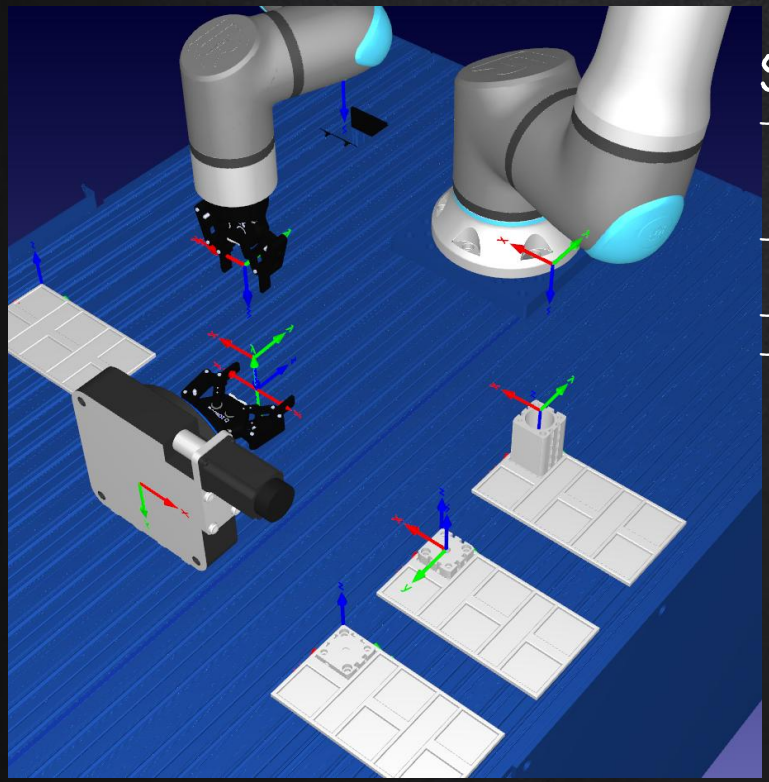
Right click on target name/Options.

Attention to active tool





Change robot configuration



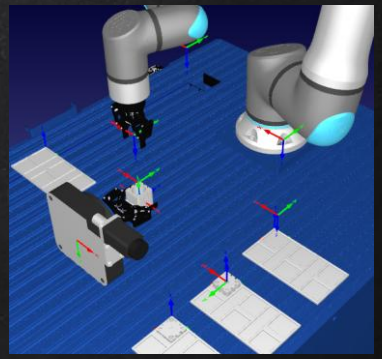
Exercice 1

Steps:

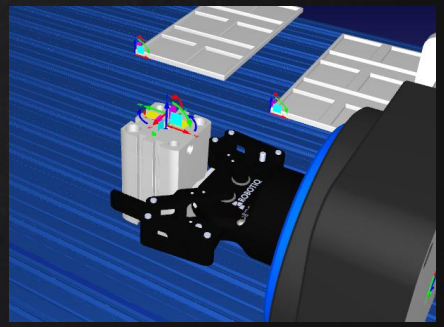
- Cylinder body psitionning on gripper 2

- Positionning down part on the body
- Rotating Gripper2
- Positionning Up part on the body

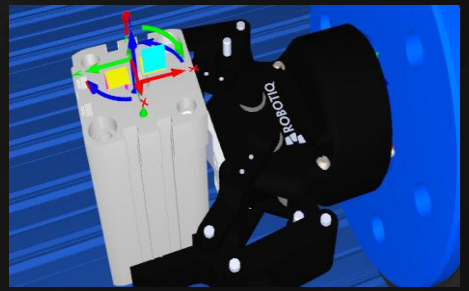
STEP 1



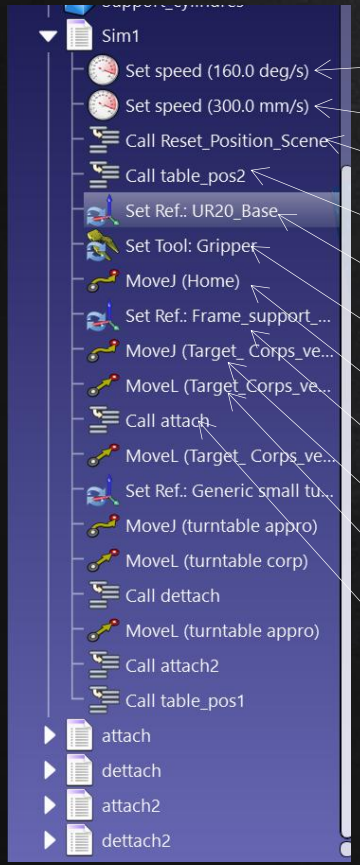
STEP 2



STEP 3



Exercice 1, steps



- Set joint velocity
- Set Cartesian velocity
- Set objects positions
- Rotate gripper 2
- Set active frame: robot base frame
- Set active tool: gripper
- Move robot to home position
- Set active frame: support
- Move robot to Target corps: type joint
- Move robot to Target corps: type linear
- Attach part (corps verin to gripper)